

Lab 5 Report: MNIST CNN Classifier

1) Baseline Comparison: MLP vs. Base CNN

The MLP baseline reached a final test accuracy of 0.9806, while the base SimpleCNN reached 0.9905. The CNN improved absolute accuracy by 0.0099.

The CNN performs better because it preserves spatial structure through convolution and pooling. MNIST digits are defined by local stroke patterns (edges, corners, curves) and their relative arrangement. A fully connected MLP first flattens each image to a 784-element vector, which removes explicit neighborhood relationships between pixels. In contrast, convolutional layers learn reusable local detectors and apply them across the image, improving parameter efficiency and translational robustness. This matches the observed performance gain.

2) Experiment Summary (MLflow Runs)

All runs used the same optimizer / training setup: learning rate = 0.001, batch size = 64, epochs = 10.

Run Name	Architecture Tag	Test Accuracy
MLP-baseline	784-256-128-10	0.9806
Simple-CNN	Conv16-Conv32-FC128-10	0.9905
CNN1	(5x5-kernel)-Conv16-Conv32-FC128-10	0.9899
CNN2	Conv32-Conv64-FC128-10	0.9908
CNN3	Conv16-Conv32-Conv64-FC128-10	0.9907

3) Analysis of Experiments (Parts 4 and 5)

Experiment A (CNN1): Larger 5x5 kernels

- Change: Swapped 3x3 kernels with 5x5 kernels in both convolutional layers.
- Result: Test accuracy = 0.9899, slightly below base CNN (0.9905).
- Interpretation: Stacking 3x3 kernels already captures the necessary detail for MNIST; larger kernels simply added unnecessary complexity

Experiment B (CNN2): Increased Filter Count

- Change: Doubled the channels from 16/32 to 32/64.
- Result: Test accuracy = 0.9908 (best run).
- Interpretation: Increasing representational capacity allowed the model to learn more diverse stroke features, providing the most meaningful performance boost.

Experiment C (CNN3): Increased Depth

- Change: Added a third convolutional layer (16 → 32 → 64).
- Result: Test accuracy = 0.9907, essentially tied with CNN2.
- Interpretation: While extra depth helps learn hierarchical features, the gains are minimal because MNIST is a relatively simple dataset.

4) Visualization Discussion

The learned first-layer filters appear as small grayscale edge/stroke detectors. Several filters emphasize vertical or horizontal transitions, while others focus on diagonals or center-surrounded intensity patterns. This is consistent with the structure of handwritten digits, which are composed of oriented strokes and curves.

Feature maps after conv1 activate strongly on local stroke fragments (such as parts of loops, edges, or intersections). Deeper feature maps (conv2/conv3) become more abstract and selective, combining earlier stroke fragments into larger digit parts. Relative to the original input, later maps are less directly human-interpretable but better encode class-discriminative patterns. This progression supports why CNNs outperform the MLP baseline on image tasks.

5) Reflection

What surprised me most was how small the differences are among strong CNN variants once accuracy is already near 99%. Changing kernel size alone did not help, while increasing channels gave the best improvement.

With another week, I would run a broader hyperparameter sweep (learning rate schedules, weight decay, dropout), add augmentation variants (small rotations/translations) to test robustness, compare training stability and generalization across seeds, and log additional MLflow metrics such as per-class accuracy and calibration error.